

86. What is $f''(x)$ equal to?

- (a) $-\frac{2}{x^3}$ (b) $2x + \frac{2}{x^3}$
(c) $6x + 3$ (d) $6x$

Consider the following data for the next **two (02)** items that follow:

Let $f(x) = |x| + 1$ and $g(x) = [x] - 1$, where $[.]$ is the greatest integer function.

$$\text{Let } h(x) = \frac{f(x)}{g(x)}$$

87. Consider the following statements:

1. $f(x)$ is differentiable for all $x < 0$
2. $g(x)$ is continuous at $x = 0.0001$
3. The derivative of $g(x)$ at $x = 2.5$ is 1

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

88. What is $\lim_{x \rightarrow 0^-} h(x) + \lim_{x \rightarrow 0^+} h(x)$ equal to?

- (a) $-\frac{3}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{1}{2}$ (d) $\frac{3}{2}$

Consider the following data for the next **two (02)** items that follow:

$$\text{Let } \phi(a) = \int_a^{a+100\pi} |\sin x| dx$$

89. What is $\phi(a)$ equal to?

- (a) 0 (b) a (c) $100a$ (d) 200

90. Which is $\phi'(a)$ equal to?

- (a) 0 (b) π (c) 100 (d) 200

Consider the following for the next **two (02)** items that follow:

A differentiable function $f(x)$ has a local maximum at $x = 0$. Let $y = 2f(x) + ax - b$.

91. Which of the following is/are correct?

1. $f'(0) = 0$
2. $f''(0) < 0$

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

92. The function y has a relative maxima at $x = 0$ for

- (a) $a > 0, b = 0$ (b) for all b and $a = 0$
(c) for all $b > 0$ only (d) for all a and $b = 0$

Consider the following for the next **two (02)** items that follow:

Let $f(x) = |x - 1|$, $g(x) = [x]$ and $h(x) = f(x)g(x)$ where $[.]$ is greatest integer function.

93. What is $\int_{-1}^0 h(x) dx$ equal to?

- (a) $-\frac{3}{2}$ (b) -1 (c) 0 (d) $\frac{1}{2}$

94. What is $\int_0^2 h(x) dx$ equal to?

- (a) $-\frac{3}{2}$ (b) -1 (c) 0 (d) $\frac{1}{2}$

Consider the following for the next **two (02)** items that follow:

$$\text{Let } \int \frac{dx}{\sqrt{x+1} - \sqrt{x-1}} = \alpha(x+1)^{\frac{3}{2}} + \beta(x-1)^{\frac{3}{2}} + c$$

95. What is the value of α ?

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) 1 (d) $\frac{4}{3}$

96. What is the value of β ?

- (a) $-\frac{2}{3}$ (b) $-\frac{1}{3}$ (c) $\frac{1}{3}$ (d) $\frac{2}{3}$

Consider the following for the next **two (02)** items that follow:

The circle $x^2 + y^2 - 2x = 0$ is partitioned by line $y = x$ in two segments. Let A_1, A_2 be the areas of major and minor segments respectively.

97. What is the value of A_1 ?

- (a) $\frac{\pi-2}{4}$ (b) $\frac{\pi+2}{4}$
(c) $\frac{3\pi-2}{4}$ (d) $\frac{3\pi+2}{4}$

98. What is the value of $\frac{2(A_1 + A_2)}{A_1 - 3A_2}$?

- (a) π (b) 1 (c) -1 (d) $-\pi$

Consider the following for the next **two (02)** items that follow:

$$\text{Let } 3f(x) + f\left(\frac{1}{x}\right) = \frac{1}{x} + 1$$

99. What is $f(x)$ equal to?

- (a) $\frac{1}{8x} - \frac{x}{8} + \frac{1}{4}$ (b) $\frac{3}{8x} - \frac{x}{8} + \frac{3}{4}$
(c) $\frac{3}{8x} + \frac{x}{8} + \frac{1}{4}$ (d) $\frac{3}{8x} - \frac{x}{8} + \frac{1}{4}$